

Long Tail Impacts of Early 2000s Subprime Loans On Today's Housing Market

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What is a Subprime Loan?

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subprime (adjective)

Subprime loans are characterized by higher interest rates, poor quality collateral, and less favorable terms in order to compensate for higher credit risk.

- We generally consider people with a FICO score of less than 670 to be subprime.
- Subprime loans usually have a rate spread of greater than 3%.

The 2008 Financial Crisis

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- Causes:
 - Subprime Loans
 - Over-leveraging
 - Poor Rating Standards
- Consequences
 - Massive Financial Downturn
 - People lost their houses
 - Banks went bankrupt



Framing the Study

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Subprime Housing Growth (Early 2000s)

Increased subprime housing expansion and credit availability.



Shift in Housing Supply

Increased production of single-family homes relative to multi-unit housing.



Lower Housing Security

Reduced long-term security due to space inefficiency and higher costs for owners and renters.

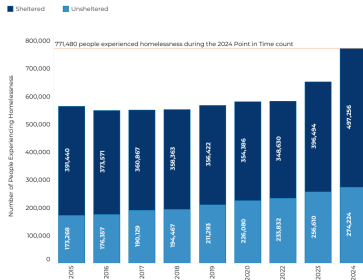
Why do we care?

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- Homelessness is increasing
- Overcrowding in homes
- Crime rate
- Poor Housing Quality

HOMELESSNESS IS ON THE RISE



Source: U.S. Department of Housing and Urban Development, 2024 AHAR: Part 1 Point-in-Time Estimates (Accessed December 2024). <https://www.huduser.gov/portal/datasets/2024-ahar-part1-1-pt-estimates-of-homelessness-in-the-us.html>
 Note: 2020 is not included due to interruptions in data collection caused by the COVID-19 pandemic.

Data Collection

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Sources

- ACS — I used the American Community Survey (5yr) for the demographic and economic data for each tract. Additionally, this was the source for the measures of housing insecurity.
- HMDA — The Home Mortgage Disclosure Act passed allows anyone to download a list of every housing loan taken out every year, including information such as loan amount, rate spread, hoepa status, etc.

Packages

- Polars / Pandas / GeoPandas
- Statsmodel / Bambi / Seaborn

Important Variables

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Dependent Variables

Housing Insecurity

- Cost Burden
- Overcrowding

Independent Variable

- Subprime Rate

Controlling Variables

- Demographic: Race, Undergraduate Percentage
- Economic: Income, Unemployment, Poverty Rate

Results — Overcrowding Rate

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	Baseline	(2)	(3)	(4)
Unemployment_PCT		-0.0002** (0.0001)		-0.0008*** (0.0001)
Poverty_Rate_PCT		0.0009*** (0.0000)		0.0002*** (0.0000)
Median_Household_Income		0.0001*** (0.0000)		0.0001*** (0.0000)
White_pct			-0.0013*** (0.0000)	-0.0013*** (0.0000)
Black_pct			-0.0012*** (0.0000)	-0.0011*** (0.0000)
Hispanic_pct			-0.0002*** (0.0000)	-0.0002*** (0.0000)
Undergrad_pct			-0.0004*** (0.0000)	-0.0005*** (0.0000)
Subprime_pct	0.1562*** (0.0048)	0.1029*** (0.0065)	0.0062 (0.0074)	0.0024 (0.0073)
Intercept	0.0138*** (0.0005)	0.0039*** (0.0012)	0.1438*** (0.0028)	0.1425*** (0.0030)
R-squared	0.0270	0.0581	0.3968	0.4044
Adj. R-squared	0.0270	0.0580	0.3967	0.4043
N	44,611	44,489	44,611	44,489

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

(2) Baseline + Economic, (3) Baseline + Demographic, (4) All Variables

Results — Cost Burden

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	Baseline	(2)	(3)	(4)
Unemployment_PCT		0.2010*** (0.0158)		-0.0043 (0.0156)
Poverty_Rate_PCT		-0.0059 (0.0075)		-0.2112*** (0.0082)
Median_Household_Income		0.0786*** (0.0021)		0.0266*** (0.0022)
White_pct			-0.2518*** (0.0065)	-0.2754*** (0.0056)
Black_pct			-0.1558*** (0.0071)	-0.1247*** (0.0064)
Hispanic_pct			-0.0795*** (0.0078)	-0.0678*** (0.0067)
Undergrad_pct			0.1269*** (0.0036)	0.0525*** (0.0039)
Subprime_pct	-36.8989*** (1.0786)	-12.3028*** (1.4482)	-19.8165*** (1.7683)	-25.1691*** (1.6327)
Intercept	20.2390*** (0.1329)	10.9132*** (0.3008)	32.2269*** (0.7134)	37.7072*** (0.6408)
R-squared	0.0291	0.0722	0.2389	0.2981
Adj. R-squared	0.0291	0.0721	0.2389	0.2980
N	44,716	44,489	44,716	44,489

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

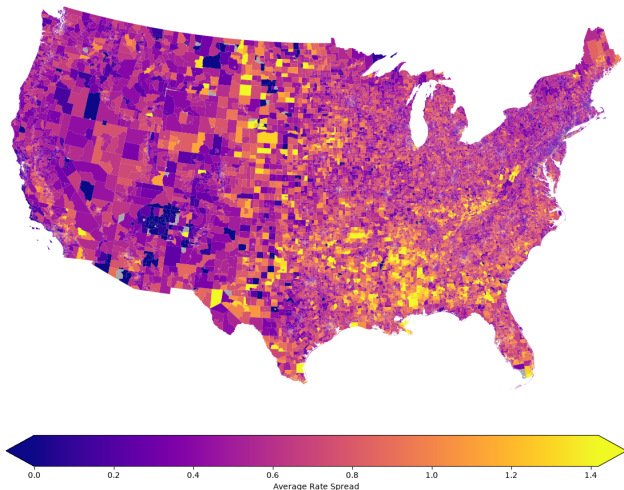
(2) Baseline + Economic, (3) Baseline + Demographic, (4) All Variables

Results — Nationwide Subprime Housing

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Average Rate Spread by US Census Tract (Continental USA)



Discussion and Conclusion

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Key Findings

- **Overcrowding & Demographics:** The relationship between subprime lending and overcrowding becomes statistically insignificant once demographic controls are added
- **Housing Cost Burden:** Higher historical subprime rates strongly correlate with lower current cost burdens across all models

Implications

Structural housing supply distortions from the early 2000s subprime expansion continue to shape tract-level housing dynamics today.